Circular Air Control Damper RD



Description
Circular air control damper with driving system.

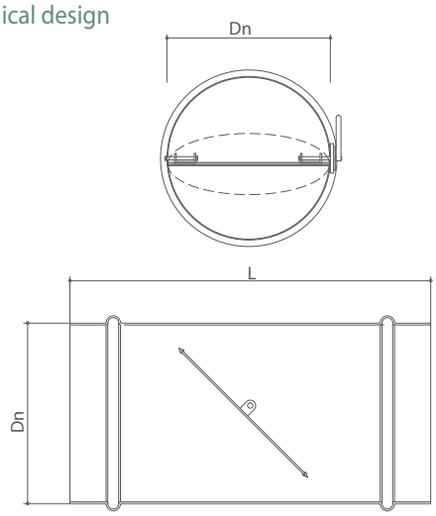
Application
To adjust the air flow in ventilation/air conditioning ducts systems.

Characteristics
The standard damper is produced with manual driving device (lever), unless electric drive (actuator) is requested.
The actuator will be selected according to the size of the damper, voltage and other technical details mentioned in the project (return spring, 0-10V control, etc.).
Actuator operates the opening/ closing of the blade through Ø8 mm shaft.
The damper is provided with a sealing gasket.

Accessories
Electric drive (actuator).

Materials and finishing
Galvanized steel sheet

Technical design



Ø	L	Dn
100	200	95
125	200	120
160	200	155
200	220	195
250	270	245
315	335	310
400	420	395

Selection table RD

Ø [mm]	Ak [mp]
100	0.0077
125	0.0121
160	0.0198
200	0.0311
250	0.0487
315	0.0774
400	0.1250

Correction factor values				
opening angle(˚)	0	15	30	45
Correction coeff.	1	1,8	8	20

ΔPt Corrected = ΔPt * Correction coeff.

Ø (mm)	Air flow m³/h	100	200	300	400	500	600	700	800	900	1000	1500	2000	3000	4000
100	NR [dB(A)] ΔPt [pa] Veff[m/s]	18 7 3.6	33 25 7.2	46 55 10.8											
125	NR [dB(A)] ΔPt [pa] Veff[m/s]		18 6 4.6	30 13 6.9	35 20 9.2										
160	NR [dB(A)] ΔPt [pa] Veff[m/s]			8 4 4.2	20 6 5.6	27 8 7.0	30 14 8.4								
200	NR [dB(A)] ΔPt [pa] Veff[m/s]				8 2 3.6	10 4 4.5	20 5 5.4	24 6.5 6.3	27 8 7.1	33 12 8.0					
250	NR [dB(A)] ΔPt [pa] Veff[m/s]						4 1 3.4	6 1.8 4.0	8 2.5 4.6	18 4 5.1	21 4.5 5.7	30 8 8.6	40 18 11.4		
315	NR [dB(A)] ΔPt [pa] Veff[m/s]											25 2.5 5.4	33 4.5 7.2	44 9 10.8	
400	NR [dB(A)] ΔPt [pa] Veff[m/s]													37 3 6.7	45 55 8.9

Annotation

* Ak [m²] - Free surface
* NR [dB(A)] - Noise level
* ΔPt [pa] - Pressure loss
* Veff[m/s] - Air velocity through the damper

Order codes

Type	+	S	/	Ø
RD		Actuator		On request